

Coding & Solution

Date	30 September 2026
Team ID	PER206AIFITTRACK
Project Name	AI-FitTrack-API
Maximum Marks	4 Marks

Coding & Solution:

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Module / File	Key Functions Implemented	Technical Implementation Details
authController.js	registerUser(), loginUser(), getUserProfile()	Validates unique email, hashes password with bcryptjs (10 salt rounds), signs JWT with HMAC-SHA256, returns user object without sensitive hash.
workoutController.js	createWorkout(), getWorkouts(), getWorkoutById(), updateWorkout(), deleteWorkout()	Performs Mongoose CRUD queries, computes estimated calorie expenditure, applies category and date range query filters, validates ownership.
aiController.js	getWorkoutRecommendation(), getFitnessInsights()	Constructs contextual prompt incorporating fitness level, goals, and equipment; calls Google Gemini 1.5 Flash API; returns structured JSON routines.
authMiddleware.js	protect()	Token extraction from Authorization header; jwt.verify() verification; attaches authenticated user document to req.user.
db.js	connectDB()	Asynchronous mongoose.connect() with connection string URI; handles graceful connection shutdown and auto-reconnect.

Code Quality & Security Verification Checklist:

S.No	Code Quality Criteria	Status	Verification Evidence
1	Strict MVC separation of concerns	Yes	Controllers, models, routes, and middleware decoupled in distinct folders
2	Environment variables used for secrets	Yes	All keys, URIs, and ports stored in .env and loaded via dotenv
3	Cryptographic password hashing	Yes	bcryptjs with 10 salt rounds applied to all user passwords
4	Stateless token authorization	Yes	Signed JSON Web Tokens validated on all protected endpoints
5	Centralized error handling	Yes	Global error middleware prevents server termination on exceptions
6	Automated test suite passing	Yes	59 automated backend test assertions passing with npm test